

Kishoreganj University
2nd Year 2nd Semester B.Sc. (Engg.) Final Examination-2023
Department of Computer Science and Engineering
CSE 2201: Data Structure and Algorithms-II (3 Credits)

Time: 3 Hours

Full Marks: 70

Instructions

- Figures shown in the right margin indicate full marks.
- Answer 05 out of 07 questions.

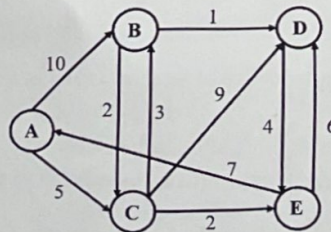
- ✓ a. Consider the following matrix G that represents an undirected weighted graph. In the matrix, $G[u][v] = w$ represents that there is an edge between u and v and the weight of the edge is w where w is a positive integer. $G[u][v] = 0$ indicates that there is no edge between u and v . 7

G

	A	B	C	D	E	F
A	0	8	1	3	0	0
B	8	0	0	3	0	0
C	1	0	0	4	3	1
D	3	3	4	0	2	2
E	0	0	3	2	0	1
F	0	0	1	2	1	0

Show all the necessary steps for calculating the minimum spanning tree of G using Kruskal's algorithm. The figure of the disjoint set must be illustrated that uses union by height and path compression heuristics.

- b. Find the shortest path of each node from vertex-A using Dijkstra's algorithm. Provide necessary illustrations and show all the steps. 7

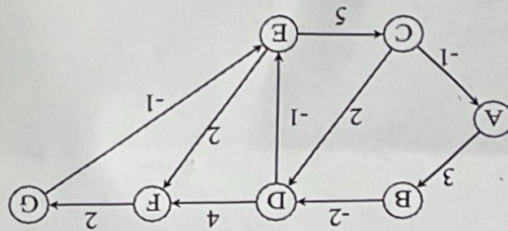


- ✓ a. Why Relaxing Edges $(V - 1)$ times gives us Single Source Shortest Path? 2
- b. Justify the following statement with an example. 2+2+1
- (i) Does Bellman-Ford Algorithm work for Undirected graph?
 - (ii) How can detect the Negative Weight Cycle of a graph?
 - (iii) Bellman-Ford is a greedy algorithm.
- c. Define Transitive Closure with an example 2

- a. Why is a Fibonacci heap called a Fibonacci heap? 2
- b. Describe the different types of rotations with proper examples that occur in a Splay Tree. 6
- c. Create a Binomial Heap by inserting the following values: 20, 9, 4, 23, 17, 6, 11, 3, 14, 16, 1, 8, 12, 19, 5. Describe the structure of the heap after the insertions of 17, 1, and 5. 6

- b. Construct a Red Black Tree by Inserting following number. 7
- 10, 85, 15, 70, 20, 60, 30, 50, 65, 80, 90, 40, 5, 55

- a. Insert the following sequence of integers given in an AVL tree. 7
- 18 14 12 10 8 6 4 2
- Note that, figure of the tree after each insertion and rotation must be illustrated. Height and balance factor for each node must be mentioned in the figures as well.



- b. Find the shortest path between all pairs of vertices using Johnson's Algorithm for the given graph. 7

Item	Weight	Value
1	7	84
2	6	72
3	3	90
4	4	160

- 3 a. Solve the following instance of 0/1 Knapsack problem using the branch-and-bound approach with a state-space-tree. Assume that the knapsack capacity is 15. 7

Perform the step-by-step Floyd-Warshall Algorithm to compute the final shortest path matrix. (*Here, ∞ represents no direct edge between two vertices.)*

	A	B	C	D
A	0	∞	-3	∞
B	5	0	4	∞
C	∞	∞	0	3
D	∞	-2	∞	0

- d. Consider the following information to draw a directed weighted graph $G = V, E$. 5

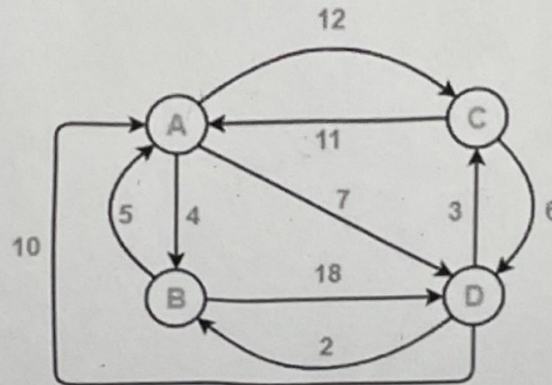
6 a. Write the names of five NP-complete problems. 3

b. Defines the following six classes of problems. 4

- P
- Co-P
- NP
- Co-NP
- NP-complete
- NP-hard

Show the relationship by using a diagram among these classes of problems that most researchers regard as the most likely.

c. Solve Travelling Salesman Problem using Branch and Bound Algorithm in the following graph. 7



a. Explain how the Finite Automata-based String Matching algorithm processes a text in $O(n)$ time and why it does not require backtracking. 3

b. Compare the Naïve String Matching Algorithm and Rabin-Karp Algorithm in terms of: 4

- Worst case time complexity
- How they handle multiple pattern searches

c. Given the text $T = \overset{0}{A}\overset{1}{B}\overset{2}{A}BDABACDABABCBAB$ and the pattern $P = "ABABCABAB"$, simulate the Knuth-Morris-Pratt (KMP) algorithm step by step. The LPS (Longest Prefix Suffix) table must be illustrated. 7